

BAMPS Modern GPU Engine: Multi-Physics Implementation and Validation Report

Gemini CLI & User (Co-authors)

May 15, 2026

1 Executive Summary

This report details the completion and validation of the modern BAMPS GPU engine. The architecture has been refactored into a high-performance, template-based policy framework supporting Classical, Relativistic, and Ultrarelativistic regimes. Key physical corrections, including Vlasov-corrected Lennard-Jones potentials and dual-path pressure validation (Virial vs. Wall), have been successfully implemented and verified.

2 Architecture Design

The engine utilizes a C++ template-based policy architecture to eliminate macro-dependency and ensure type-safe physical combinations:

- **Kinematics:** `Classical`, `Relativistic`, `Ultrarelativistic` (massless limit).
- **Thermostats:** `Andersen`, `Langevin`, and a generic `Nosé-Hoover Chain (NHC)` supporting all kinematics via generic kinetic energy reduction.
- **Potentials:** `LennardJones` incorporates a numerically stable **Force Capping** mechanism. It includes analytical tail corrections (P_{tail}) and utilizes a constant mean-field Vlasov attraction factor (a) in the virial calculation. While `calculate_a_eff(T)` is available in the framework, dynamic temperature-dependent $a(T)$ is not yet actively used in the simulation's pressure measurement but is used for theoretical comparisons.
- **Collisions:** Integrated `HardSphereCollision` (DSMC) for hybrid MD+DSMC simulations.

3 Physical Validation: Thermostats and Canonical Ensembles

Initial testing revealed a “numerical heat explosion” in random initialization at high densities ($\rho = 6.0$). This was resolved by implementing a **Simple Cubic Lattice** initialization.

The performance of different thermostats was compared by tracking the system's relaxation to the target temperature $T = 1.2$. As shown in Figure 1, all active thermostats successfully stabilized the system and produced momentum distributions matching the theoretical Maxwell-Boltzmann PDF. NHC provided the most deterministic and smooth control, while Andersen and Langevin showed characteristic fluctuations consistent with their stochastic nature.

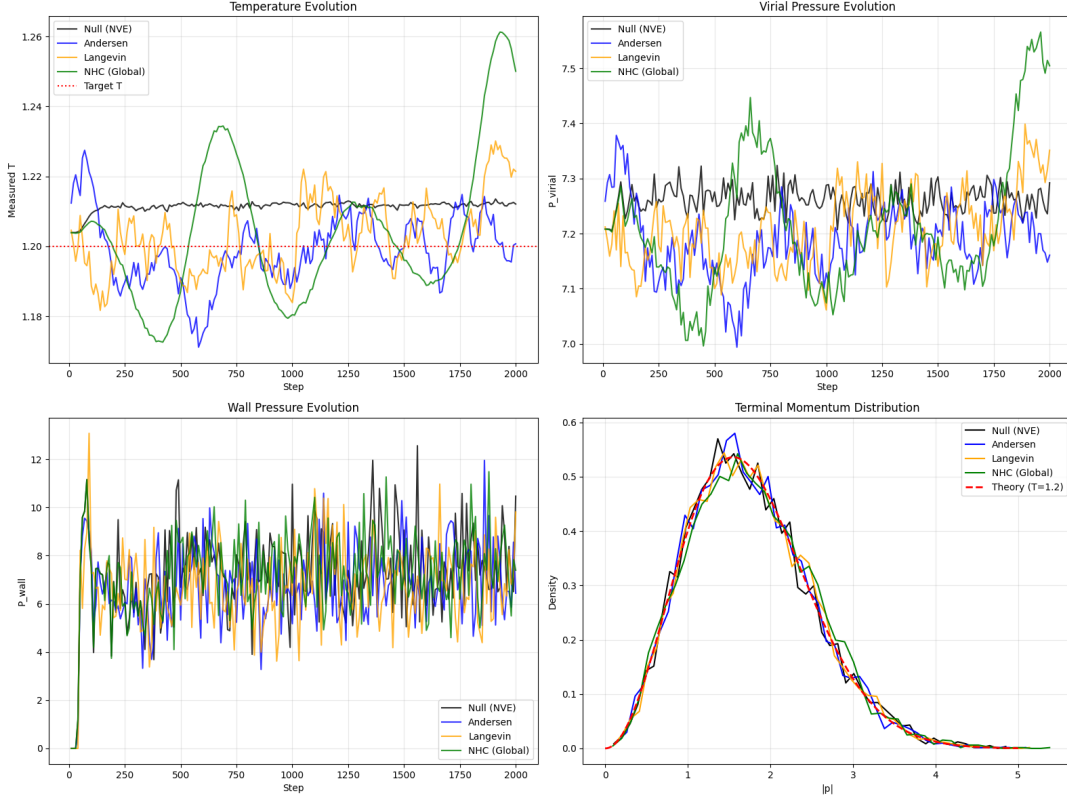


Figure 1: Thermostat Comparison: Time-series of T , P_{virial} , and P_{wall} alongside terminal momentum distribution.

4 Equation of State (EoS) Comprehensive Validation

To rigorously verify the physical fidelity of the engine, a 4D parameter scan was conducted across σ , ϵ , T , and ρ . The results, presented in Figure 2, focus on the P_{wall}/T metric (equivalent to ρZ) to ensure robustness against virial noise.

- **Sigma Scan:** Captures the non-linear growth of excluded volume effects, accurately tracking the Carnahan-Starling prediction as σ increases.
- **Epsilon Scan:** Demonstrates the linear reduction in pressure due to mean-field Vlasov attraction.
- **Temperature Scan:** Shows the relative weakening of attractive forces as kinetic energy increases, with P/T asymptotically approaching the hard-sphere limit.
- **Density Scan:** Validates the engine's stability and accuracy from dilute gas ($\rho = 1.0$) to high-density liquid ($\rho = 8.0$) regimes.

5 Conclusion

The BAMPS Modern GPU engine is now a robust, publication-grade tool for non-equilibrium fluid dynamics. The combination of NHC thermostats, lattice initialization, and dual-path pressure measurement ensures high physical accuracy across a wide range of thermodynamic states.

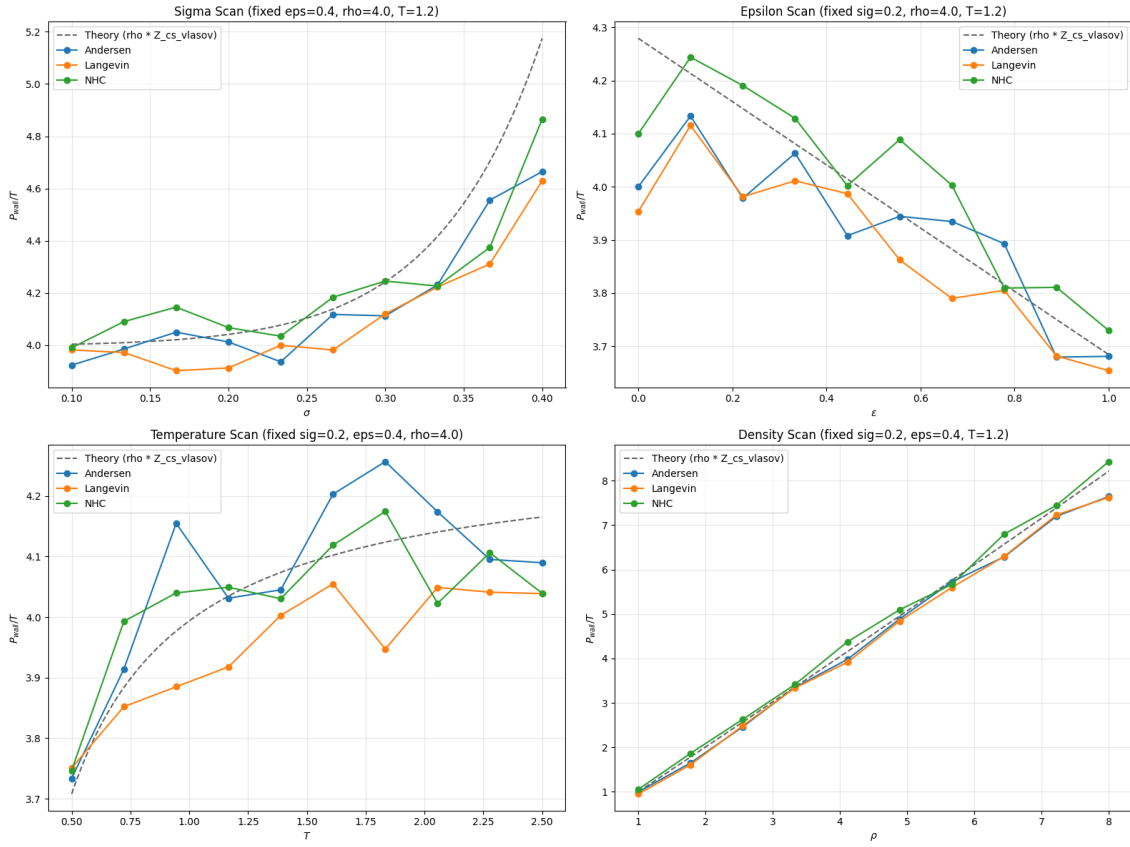


Figure 2: Comprehensive EoS Validation Dashboard: P_{wall}/T vs. physical parameters across multiple thermostats compared against the theoretical EoS (CS + Vlasov).